

POINT IROQUOIS, MI, USA

WMO: 997265

Lat: 46.4846N

Lon: 84.6309W

Elev: 186

StdP: 99.11

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.6	-20.0	N/A	N/A	N/A	N/A	N/A	N/A	21.2	-5.8	17.0	-4.4	3.8	120	0.646

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	27.7	N/A	25.6	N/A	23.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.8	11.0	DB	-26.7	30.3	4.4	1.6	-29.9	31.5	-32.5	32.4	-34.9	33.3	-38.1	34.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-8.8	-8.7	-4.3	2.7	9.0	14.6	18.1	18.2	14.7	8.5	2.0	-4.0	(6)	
	DBStd	10.66	6.24	5.80	5.83	3.99	4.40	3.79	3.28	2.63	3.63	4.06	4.53	5.05	(7)	
	HDD10.0	2607	582	524	442	222	72	5	0	0	4	79	242	435	(8)	
	HDD18.3	4869	840	758	700	469	292	122	45	34	120	306	491	693	(9)	
	CDD10.0	871	0	0	1	2	41	143	251	255	143	32	2	0	(10)	
	CDD18.3	92	0	0	0	0	3	11	38	31	9	1	0	0	(11)	
	CDH23.3	462	0	0	0	1	34	92	182	112	38	3	0	0	(12)	
	CDH26.7	91	0	0	0	0	9	22	33	20	7	0	0	0	(13)	
(14) Wind	WSAvg	4.5	5.0	4.6	5.0	5.1	4.5	3.7	3.3	3.7	3.9	4.6	4.8	5.5	(14)	
(15) Precipitation	PrecAvg	875	58	40	52	64	68	77	79	77	107	97	86	69	(15)	
	PrecMax	1162	113	85	123	125	130	158	170	118	227	172	176	129	(16)	
	PrecMin	619	23	6	5	29	18	21	29	36	19	27	34	5	(17)	
	PrecStd	141	22	20	28	28	28	29	36	24	47	41	35	32	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	5.8	12.2	18.7	28.0	29.8	29.8	29.1	27.5	22.7	15.7	6.9	(19)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)	
	2%	DB	3.1	1.8	8.2	13.4	22.3	26.2	27.5	26.5	23.9	19.0	11.5	5.0	(21)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)	
	5%	DB	1.8	0.5	5.2	10.2	18.4	23.0	25.5	24.3	21.7	16.4	9.3	3.6	(23)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)	
	10%	DB	0.3	-0.7	2.9	7.6	15.4	20.3	23.4	22.3	19.9	14.2	7.8	2.6	(25)	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)	
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.5	8.0	6.4	8.1	7.9	7.4	6.3	7.1	5.6	5.3	5.5	(35)	
		MCDBR	5.8	7.4	9.8	11.8	14.3	12.9	11.6	10.1	10.0	8.7	7.2	5.8	(36)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.281	0.291	0.319	0.355	0.371	0.389	0.390	0.378	0.367	0.332	0.303	0.286	(40)	
	taud		2.344	2.302	2.334	2.332	2.374	2.346	2.365	2.399	2.424	2.503	2.509	2.382	(41)	
	Ebn at Noon		816	882	907	900	895	876	870	867	846	830	794	767	(42)	
	Edh at Noon		73	94	106	118	117	121	117	109	98	78	64	64	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.07	1.99	3.25	4.45	5.56	5.99	6.03	5.10	3.80	2.21	1.15	0.83	(44)	
	RadStd		0.11	0.21	0.37	0.51	0.45	0.40	0.34	0.34	0.28	0.25	0.15	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page